What has led to the rise of agent-based systems?

Over the past decades, manufacturing has evolved within an increasingly globalized and competitive economy, where customers demand highly customized, high-quality products at lower costs. To remain competitive, companies must increase flexibility while maintaining quality. Agent-based systems have emerged as a promising approach to address these challenges (Leitão, 2009). Among the various application domains of agent technology, manufacturing is considered one of the most significant (Luck, 2005).

Traditional manufacturing control systems are typically based on centralized architectures due to the complexity of production environments, which involve numerous interactions and functions. Although these systems can achieve high productivity, they often rely on large monolithic software solutions that require extensive customization and maintenance (Leitão, 2009).

To overcome these limitations, researchers have proposed intelligent and distributed control systems in which complex problems are decomposed into smaller tasks assigned to autonomous agents. Each agent possesses its own objectives, knowledge, and capabilities, while global decisions emerge through collaboration among multiple agents. This paradigm is known as a Multi-Agent System (MAS), characterized by autonomy, adaptability, decentralization, and scalability (Leitao and Colombo, 2006).

Beyond control systems, MAS has been applied successfully to production planning, scheduling, logistics and supply chain management, energy and emission reduction, and decision support systems (Pulikottil *et al.*, 2021). The diversity of these applications demonstrates the ability of MAS to support both operational decision-making and strategic optimization across manufacturing systems.

Despite its potential, the industrial adoption of MAS remains limited (Marik and McFarlane, 2005). Technical barriers include the lack of a general reference architecture and the difficulty of integrating heterogeneous production systems, protocols, and standards (Pulikottil *et al.*, 2021, 2023). Organizational barriers also exist, as the decentralized nature of MAS may reduce acceptance among managers accustomed to centralized decision-making structures (Leitão, 2009). Overcoming these challenges is essential to realizing the full potential of MAS in manufacturing.

To sum up, Multi-Agent Systems provide a flexible, distributed alternative to traditional centralized manufacturing control systems, with applications in planning, scheduling, logistics, and decision-making. However, their industrial adoption remains limited by technical and organizational challenges.

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